PART 3

Cardiac Emergencies and Resuscitation

IN THIS PART

1. Describe the pumping and electrical functions of the heart in plain terms.
2. Recognize acute chest pain patterns and acute coronary syndrome, and state first priorities.
3. Recognize a short list of dangerous rhythm patterns and what they mean for the patient.
4. Describe high-quality adult CPR and the use of an automated external defibrillator (AED), step by step.
5. Recognize shock and state the common first priorities.

Some patients arrive with a warning: a pressure in the chest, a sweat that does not fit the weather, a sense that something is wrong. Others arrive with no warning at all, collapsed on a floor. Both stories live in the heart, and this Part follows them in order, from the quiet signs of a heart in trouble to the loud emergency of a heart that has stopped.

The Part begins with the heart itself, because the signs make sense only once you know what the organ does. It then covers chest pain and acute coronary syndrome, a common emergency that threatens the heart muscle, followed by a short guide to the rhythms you may see on a monitor or be told about. The longest entry is cardiac

arrest, high-quality CPR and the AED, since these skills matter in almost every setting where you may work. The Part closes with shock, the state in which the circulation fails to supply the body, whatever the cause.

As in every Part, what you may do depends on your scope of practice and local protocol. Where a task is outside your role, the right action is to call for help and begin what you are trained to do while you wait.

3.1 The Heart in Brief: Pump, Vessels and Electrical System

The heart is a muscular pump about the size of a closed fist, lying in the chest between the lungs. It does two jobs. It moves blood to the lungs to collect oxygen, and it moves oxygen-rich blood to every organ, including the heart muscle itself.

WHAT IT IS. The heart has four chambers. The two upper chambers, the atria, receive blood. The two lower chambers, the ventricles, push it out. The right side sends blood to the lungs, and the left side sends it to the rest of the body. Valves between the chambers keep blood moving in one direction. The heart muscle has its own blood supply through the coronary arteries, which run over its surface. If one of them becomes blocked, the part of the muscle it feeds is starved of oxygen.

Figure 9 shows the four chambers and the two coronary arteries; notice that the arteries run over the outside of the muscle they feed.

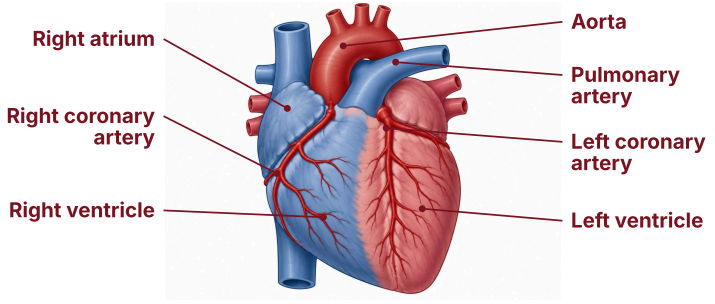


Figure 9. The heart and its coronary arteries, seen from the front.

THE ELECTRICAL SYSTEM. Each heartbeat is triggered by a small electrical signal that starts in the upper chambers and spreads through the muscle in a set order, so that the chambers squeeze in sequence. When the signal is disturbed, the heart may beat too fast, too slow, irregularly, or not effectively at all. A monitor draws that signal as a line, and Entry 3.3 explains what a few of the patterns mean.

HEART FAILURE. The heart can also weaken over time, a condition called heart failure, in which the pump no longer moves blood as well as it should. Patients are often short of breath, tire easily and may have swollen ankles. A flare-up can make breathing hard, particularly when lying flat, so the patient may be more comfortable sitting upright. Treat a worsening breathing problem as an emergency, sit the patient upright, call for help and use the distress priorities of Entry 2.4.

WHY IT MATTERS. A responder who knows that the heart both pumps and conducts can reason about what fails. A blocked coronary artery damages the muscle. A disturbed signal changes the

rhythm. A weak pump lowers the flow of blood.

NURSING AND RESPONDER PRIORITIES. Keep the two functions in mind, the pump and the electrical signal, and ask of any cardiac problem which of them has failed.

See also: Entry 3.2 (chest pain); Entry 3.3 (rhythms); Entry 3.4 (cardiac arrest).

3.2 Acute Chest Pain and Acute Coronary Syndrome

After injuries, chest pain is the second most common reason that adults come to a United States emergency department, and most chest pain turns out not to be a heart attack. The difficulty is that a heart attack can look like many harmless things, so the safe approach treats every serious chest pain as cardiac until a clinician shows otherwise.

WHAT IT IS. Acute coronary syndrome, written ACS, is the name for a group of conditions in which blood flow to part of the heart muscle drops suddenly. The best-known is the heart attack, or myocardial infarction (MI), in which a section of muscle begins to die. Cardiac arrest is a different event: it is the stopping of an effective heartbeat, and it may follow a heart attack, but the two are not the same.

HOW A HEART ATTACK DEVELOPS. Most heart attacks begin when a fatty deposit in a coronary artery cracks and a blood clot forms on it, cutting off flow. The longer the muscle goes without blood, the more of it dies, which is why the call for help comes before any further assessment.

RECOGNITION. The classic picture is pressure, tightness or squeezing in the center of the chest that may spread to the arm, jaw, neck, back or upper abdomen. It may come with shortness of breath, sweating, nausea, light-headedness or a sense of dread. It may start slowly or suddenly, may come and go, and does not ease with rest. The picture is not always classic. Women, older adults and people with diabetes more often have less typical symptoms, such as shortness of breath, nausea, unusual tiredness or discomfort in the back or jaw, with little or no chest pain. A patient who looks calm and says "it's probably indigestion" may still be having a heart attack.

RED FLAGS

Chest pressure or pain lasting more than a few minutes or returning, with shortness of breath, sweating, nausea, faintness, a pale or gray color, or collapse. Call emergency services at once and do not let the patient drive.

WHO IS AT HIGHER RISK. Risk rises with age, smoking, high blood pressure, high cholesterol, diabetes, being overweight, little physical activity and a family history of heart disease. A patient with several of these deserves extra caution, but a patient with none can still have a heart attack, so a risk list is no reason to hold back.

ANGINA AND HEART ATTACK. Angina is chest discomfort that appears when the heart muscle needs more oxygen than a narrowed artery can deliver, usually during effort or stress, and it eases within minutes of rest or of the patient's prescribed treatment. A heart attack happens when a blocked artery stops blood flow long enough for heart muscle to begin to die, so the discomfort does not settle with rest. You will not always know which you are looking at. A patient with known angina whose pain is

stronger than usual, comes on at rest, or lasts longer than their usual attacks should be treated as having a possible heart attack and given the same urgent response.

WHAT TO ASK. Use short, direct questions. When did it start? What were you doing? What does it feel like, and does it spread? Is it constant or does it come and go? Have you had it before, and is it different this time? Do you have heart disease, high blood pressure, diabetes or allergies? Which medicines do you take, and have you taken anything for this? Write the answers down with the times, because the next team will want them and the patient may not be able to repeat them.

OTHER CAUSES. Chest pain also arises from the lungs, the muscles and bones, the food pipe and the large vessels. Some of these are as dangerous as a heart attack. A sudden tearing pain that spreads to the back, pain with a sudden shortness of breath, or pain after an injury deserves the same urgent response. You do not have to decide the cause. Your task is to recognize the pattern that needs urgent help and to act on it.

FIRST PRIORITIES. Call emergency services. Help the patient rest in the position that is most comfortable and keep them still and reassured. Check airway, breathing and circulation and take vital signs. Use the questions above, and ask about allergies before any aspirin is offered. Where protocol allows, a patient who has no allergy to aspirin and has not been told to avoid it may be helped to chew aspirin. Giving medicines is outside many roles; follow your scope and medical direction. Do not give anything by mouth if the patient is faint or has trouble swallowing. Be ready to begin CPR and to use an AED if the patient collapses and stops breathing normally.

SCOPE NOTE

Decisions about medicines for chest pain belong to clinicians acting under medical direction. A student or lay responder calls for help, positions and reassures the patient, monitors, and follows protocol if instructed to assist.

REASSESSMENT. A patient with chest pain can change in minutes. Ask again about the pain at short intervals and record the time and the answer. Repeat the vital signs and note the color and sweating. Report any change in the pain, new breathlessness, faintness or a change in the pulse at once, since each may mean the heart is under greater strain. A patient who was comfortable and becomes pale and sweaty has deteriorated, whatever the numbers say.

NURSING AND RESPONDER PRIORITIES. Treat serious chest pain as cardiac, call early, keep the patient still, monitor closely for deterioration, and be ready for cardiac arrest.

See also: Entry 1.3 (primary survey); Entry 3.4 (cardiac arrest); Entry 6.4 (chest pain in older adults).

3.3 Recognizing Dangerous Rhythms

A heart monitor draws the electrical signal of the heart as a moving line. You do not need to interpret it as a specialist would. You need to recognize a small number of patterns that tell you how worried to be, and to know what to do when a monitor shows one.

WHAT IT IS. A rhythm is the pattern of the heart's electrical activity over time. The normal pattern shows a regular series of small and large waves with a steady rate. Abnormal rhythms may

be too fast, too slow, irregular, or chaotic. A rhythm is dangerous when it prevents the heart from filling or emptying, so that blood pressure falls and the brain and heart go short of oxygen.

WHY IT MATTERS. A monitor tracing is only a picture of electricity, and it can mislead. A patient can have a normal tracing and no effective pulse, and a patient can have an abnormal tracing from a loose lead or movement and feel quite well. Read the tracing together with the patient's responsiveness, breathing and skin color.

Rhythm patterns and what to do

Pattern	What it looks like	What it means
Regular rhythm	Even beats at a steady spacing	Usually reassuring if the patient is well
Fast or irregular rhythm	Rapid or uneven beats	May cause faintness or chest pain; call for help
Chaotic rhythm	A wavy, disorganized line with no clear beats	The heart is quivering and not pumping; if the patient is also unresponsive and not breathing normally, this is cardiac arrest
Slow rhythm	Widely spaced beats	May cause faintness, confusion or collapse; call for help if the patient has symptoms
Flat line	No electrical activity	Check the leads and the patient, then treat as cardiac arrest if the patient is unresponsive

Figure 10 shows four of these patterns, simplified; notice that the regular line differs from the chaotic one in spacing as well as in shape.

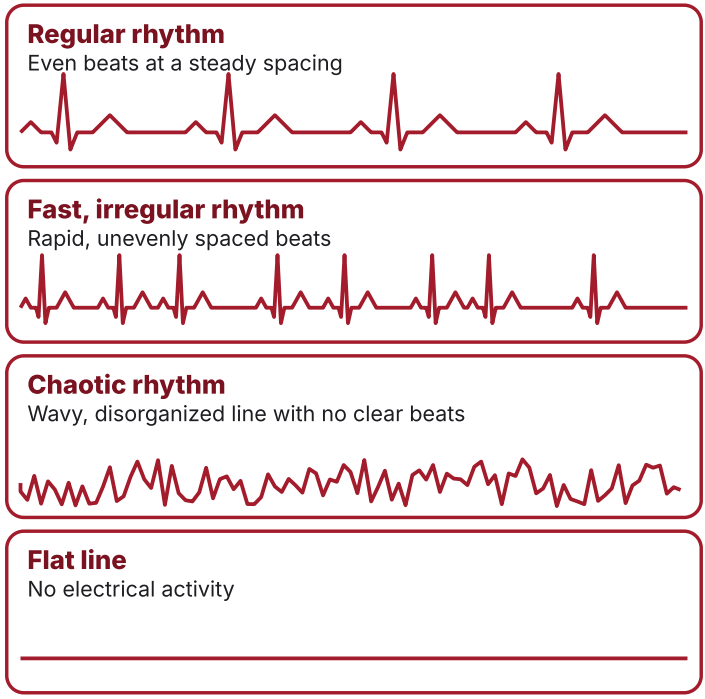


Figure 10. Simplified drawings of a regular rhythm, a fast irregular rhythm, a chaotic rhythm and a flat line.

TAKING A PULSE. Entry 1.4 described the pulse check. With a monitor attached, compare your count with the monitor's rate; if the two do not match, report the difference and trust your assessment of the patient. If you are a trained health care professional, check for a pulse at the neck of an unresponsive patient for no more than 10 seconds; if you are not sure you feel one, begin compressions.

NAMING THE PATTERNS. Clinicians give these patterns names. The chaotic pattern is usually ventricular fibrillation, and a flat line is called asystole. A very fast, regular pattern can also stop the heart from pumping, which is why a fast rhythm in a faint patient needs urgent help. An AED can shock a chaotic pattern but does not shock a flat line, so when it advises no shock you keep doing compressions. A slow rhythm can be as dangerous as a fast one because too little blood leaves the heart. Atrial fibrillation, a common irregular rhythm in older adults, is not always an emergency by itself, but you report it and watch the patient.

WHAT TO DO. Look at the patient first. A flat line on a monitor in a patient who is awake and talking is almost certainly a loose lead or an equipment problem, so check the lead and the monitor. A chaotic pattern in an unresponsive patient who is not breathing normally is cardiac arrest, and the response is CPR and defibrillation as soon as a defibrillator is ready, as Entry 3.4 describes. A fast or irregular rhythm in a patient who is faint, has chest pain, or is short of breath needs urgent help from a clinician. Report what you see: the rate, whether the beats are regular, and how the patient looks.

RED FLAGS

Faintness, chest pain, shortness of breath or confusion with a fast, slow or irregular rhythm, or any monitor change in a patient who is unresponsive and not breathing normally. Call for help at once.

NURSING AND RESPONDER PRIORITIES. Read the patient before the screen, and start CPR for an unresponsive patient who is not breathing normally.

See also: Entry 3.1 (electrical system); Entry 3.4 (CPR and AED).

3.4 Cardiac Arrest, High-Quality CPR and the AED

Cardiac arrest is the sudden stop of effective heartbeat, and it is the emergency in which a bystander's actions matter most. A patient in cardiac arrest is unresponsive and is not breathing normally. Blood stops flowing to the brain, and the patient collapses and loses consciousness. Chest compressions keep a little blood moving, and the AED is the tool that can correct a dangerous rhythm so that the heart can resume pumping.

WHAT IT IS. Cardiac arrest happens when the heart stops pumping blood effectively. Causes include a heart attack, a dangerous rhythm, severe bleeding, a blocked airway, drowning, poisoning and injury. It is different from a heart attack, in which the heart usually keeps beating. Cardiopulmonary resuscitation (CPR) combines chest compressions with rescue breaths. An AED is a portable device that analyzes the heart rhythm and, if it finds a rhythm that can be corrected by a shock, tells the rescuer to deliver one.

WHY IT MATTERS. Early CPR and prompt defibrillation are the most important interventions for survival, and a bystander decides how soon they begin. Every minute without CPR and defibrillation lowers the chance of survival, which is why the first person on the scene must start at once, before the cause is known. Good compressions are hard work and can be done poorly, so quality and the swapping of rescuers matter as much as speed of starting.

RECOGNITION. Check for a response by tapping the shoulders and shouting. Look for breathing for no more than 10 seconds. Occasional gasps are not normal breathing and are a sign of cardiac arrest. If the patient is unresponsive and is not breathing, or

is only gasping, treat it as cardiac arrest. If you are a trained health care professional, you may check for a pulse during the same 10 seconds; if you are not sure you feel one, begin compressions.

RED FLAGS

Unresponsive, no breathing or only occasional gasps, no response to voice or touch, a gray or blue color, or a collapse with no warning. Call emergency services, send for an AED and begin compressions at once.

THE SEQUENCE. Press hard and fast, at a rate of 100 to 120 compressions a minute, to a depth of at least 2 inches (5 centimeters) but not more than about 2.4 inches (6 centimeters) in an adult, and let the chest return fully between compressions. Keep interruptions as short as you can. If you are trained and willing to give breaths, give 2 breaths after every 30 compressions, each lasting about 1 second and just enough to make the chest rise. If you are not trained, continue compressions alone. If you are alone, put the phone on speaker, call emergency services, begin compressions, and use an AED as soon as one is within reach, as your training directs.

Figure 11 shows the hand and body position.

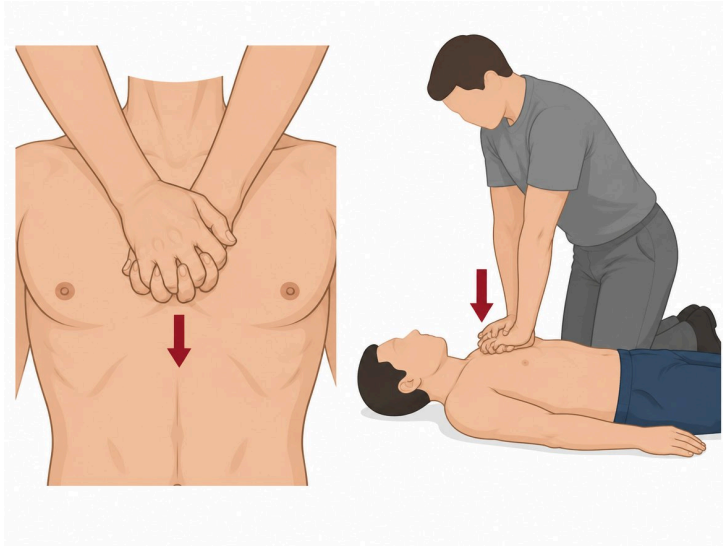


Figure 11. The hand position and the straight-armed body position for chest compressions on an adult.

PROCEDURE: ADULT CPR WITH AN AED

Purpose: to keep blood moving and to deliver a shock when the AED advises one. Scope: follows your training level and local protocol.

1. Check that the scene is safe, tap the shoulders and shout; if there is no response, call for help.
2. Send someone for an AED and ask them to call emergency services.
3. Look for breathing for no more than 10 seconds; if the patient is not breathing normally, begin compressions.

4. Place the heel of one hand on the lower half of the breastbone, with the other hand on top, arms straight.
5. Press at 100 to 120 a minute to a depth of at least 2 inches (5 centimeters), letting the chest rise fully each time; if you are trained, give 2 breaths after every 30 compressions.
6. When the AED arrives, turn it on and follow its voice prompts; attach the pads as shown on them.
7. Make sure no one is touching the patient while the AED analyzes and again before a shock is given.
8. Resume compressions immediately after the shock or if no shock is advised, and swap rescuers about every 2 minutes.
9. Continue until emergency responders take over, the patient moves or breathes normally, or you are too exhausted to continue.

> **⚠ SAFETY**
>
> Do not touch the patient while the AED analyzes or delivers a shock. Dry a wet chest and move the patient away from standing water before attaching the pads. Do not delay compressions to look for the cause.

CALL FOR HELP AT ONCE IF the patient is unresponsive and not breathing normally, and ask someone to take over if you cannot continue.

Figure 12 shows the whole sequence as one flow, from collapse to handover.

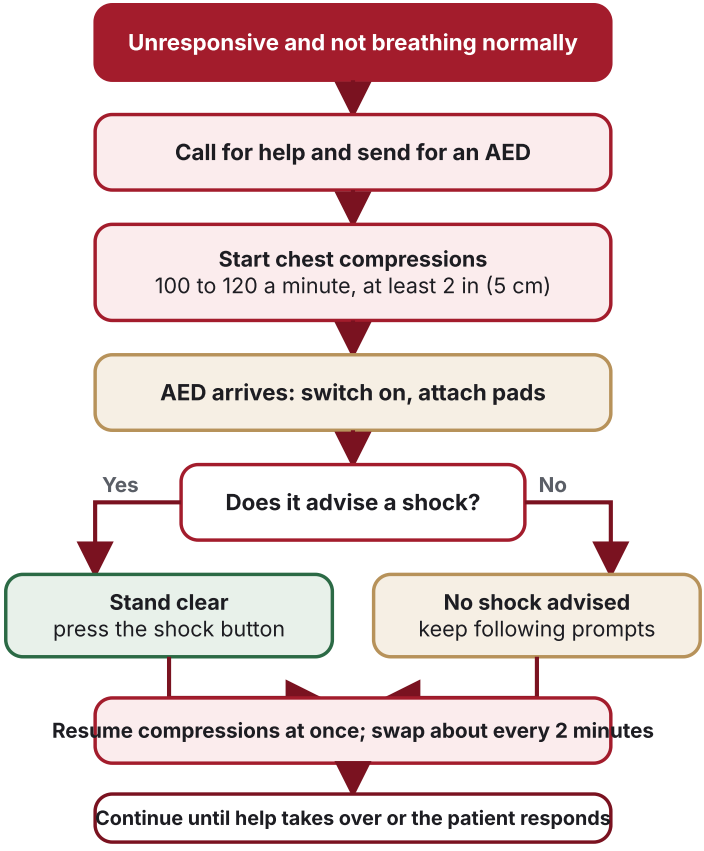


Figure 12. The adult cardiac arrest response from recognition to handover.

The steps follow the physiology. Compressions create a small flow of blood to the brain and heart, so they begin first and are interrupted as little as possible. The AED is used as soon as it arrives, and compressions continue until it is switched on, the pads are at-

tached and it is ready to analyze the rhythm. Swapping rescuers about every 2 minutes keeps the quality of compressions from falling as arms tire.

USING THE AED. An AED is designed for people with no training. When you open it or switch it on, it talks you through every step. Bare the chest, dry it, and place the two sticky pads as the picture on the pads shows. The device then reads the rhythm. Do not touch the patient while it does. If it advises a shock, check that everyone is clear and press the shock button. If no shock is advised, resume compressions. Either way, leave the pads on and keep following its prompts. For AED use, adult pads and the adult steps are used for a child aged about 8 years or older. Entry 6.2 covers CPR and AED differences in infants and younger children.

> **REMEMBER**
>
> Hard, fast, clear and continuous: press hard, press fast, let the chest rise, and keep interruptions short. If the patient is unresponsive and you are unsure whether the breathing is normal, start compressions.

COMPRESSION QUALITY. Compressions are weakened by pressing too shallowly or too deeply, pressing too slowly, too fast or unevenly, and leaning on the chest between compressions so that it cannot recoil. Keep your shoulders over your hands and your arms straight so that the weight of your upper body does the work. A firm surface under the patient helps. Counting aloud keeps the rate steady. Compressions can break ribs, and this is not a reason to stop; the priority is to keep blood moving.

SPECIAL SITUATIONS WITH THE AED. Remove any medication patch from the chest, with a gloved hand if you have one, and wipe the skin before attaching the pads. Do not place a pad directly over a lump under the skin where a pacemaker or similar

device sits; place it a short distance away. Chest hair rarely stops pads from sticking, so do not delay the shock to remove it. If the patient is a child younger than about 8 years, use pads or a setting for children if the device has them, and use the adult pads if it does not. Follow the device's voice prompts first.

HANDS-ONLY CPR. A bystander who has no training, or who is unwilling or unable to give breaths, can still help by giving continuous chest compressions at the rate and depth described above until the AED arrives or help takes over. Compressions alone are far better than waiting. A trained responder gives breaths as described, particularly for patients whose arrest follows drowning, a drug overdose or a breathing problem.

AFTER THE PATIENT RESPONDS. If the patient begins to breathe normally and responds, keep the airway open, keep the patient warm and still, leave the AED pads in place, and keep watching. Keep checking that the patient is still breathing normally. Be ready to resume compressions at once and report to the paramedics exactly what you did and when.

PREGNANCY. In late pregnancy, the enlarged uterus can press on the large vessels and reduce the blood returning to the heart. Entry 6.3 describes the adaptation. Begin CPR without delay; Entry 6.3 describes how the uterus is moved to the left by hand while compressions continue.

NURSING AND RESPONDER PRIORITIES. Recognize the arrest at once, call for help and an AED, press hard and fast with few interruptions, and follow the AED until help takes over.

See also: Entry 1.3 (primary survey); Entry 2.2 (choking); Entry 6.2 (child and infant); Entry 6.3 (pregnancy).

3.5 Shock: Recognition and First Priorities

Shock is not an emotional state. It is a failure of the circulation to deliver enough blood and oxygen to the organs, and it can have several causes. It is hard to spot in its early stages because the body compensates, and a patient who is in shock may look calm for a time.

WHAT IT IS. The circulation needs four things: enough blood volume, a pump that works, vessels that are the right size, and a clear path for the blood to flow. Shock occurs when one of them fails. The blood volume may fall from bleeding or fluid loss. The pump may fail, as in a heart attack. The vessels may widen abnormally, as in a severe infection or an allergic reaction. Something may also block the flow, as with air trapped around a lung or fluid around the heart. In each case the organs receive too little, and without treatment the failure spreads.

FOUR BROAD TYPES. Grouping shock by what has failed helps you recognize the cause and anticipate the first actions.

Four types of shock

Type	What has failed	Examples
Low blood volume	Too little blood or fluid in the circulation	Severe bleeding, burns, severe vomiting or diarrhea
Pump failure	The heart cannot move blood effectively	Heart attack, dangerous rhythm
Widened vessels	Vessels dilate and blood pressure drops	Severe infection, anaphylaxis
Blocked flow	Something prevents blood from moving	Collapsed lung under pressure, fluid around the heart

WHY IT MATTERS. The body protects the brain and heart by narrowing the vessels elsewhere and speeding the heart. In this early, compensated stage the blood pressure may stay normal, so a normal pressure does not exclude shock. When the compensation runs out, the stage is called decompensated, the pressure falls suddenly and the patient can deteriorate within minutes. The aim is to recognize shock before that point.

EARLY ACTIONS BY TYPE. The first actions depend on the cause. For heavy bleeding, control it (Entry 4.2). For anaphylaxis, call for help and, where protocol allows, help with the patient's own epinephrine auto-injector. Giving medicines is outside many roles; follow your scope and medical direction. For a suspected heart cause, call for help, keep the patient still and be ready to start CPR. For a suspected infection, call for help and report the signs, because early treatment is given by clinicians. In every case,

keep the patient warm and recheck often. Lay the patient flat unless breathing is difficult or the cause is a heart problem or chest pain, when the most comfortable position is safer.

RECOGNITION. Look for a fast heart rate, a fast breathing rate, cool, pale or mottled skin, a slow capillary refill, a weak pulse, new confusion, anxiety or drowsiness, and thirst. The pressure may be normal early and low later. Ask what has happened: bleeding, an injury, vomiting, an infection, an allergic reaction or a heart problem. Different causes lead to different first actions, which are described in their home entries.

RED FLAGS

Cool, pale or mottled skin, a fast weak pulse, fast breathing, new confusion or drowsiness, a low blood pressure, or a patient who looks worse than the numbers suggest. Call for help at once.

FIRST PRIORITIES. Call for help. Control any visible severe bleeding first, if your protocol directs it. Lay the patient flat unless they are struggling to breathe or have chest pain, and in that case let them take the most comfortable position; keep them warm, and do not give food or drink. Give oxygen as your protocol directs. Giving medicines is outside many roles; follow your scope and medical direction. Recheck the vital signs at short intervals and write down the times. Entry 4.2 covers bleeding control, Entry 2.5 covers anaphylaxis, and Entry 5.4 covers sepsis.

REASSESSING SHOCK. Shock changes quickly, so recheck at short intervals. A heart rate that is falling toward normal, a rising blood pressure, warmer skin and clearer thinking suggest the patient is responding. A rising heart rate, a falling blood pressure, cooler skin and more drowsiness suggest the patient is worse, and

you should tell the clinician at once. Record each set of observations with the time, so that the trend is visible to the team that takes over.

NURSING AND RESPONDER PRIORITIES. Recognize shock from the whole pattern as well as the pressure, call for help, treat the cause you can see, and reassess often.

See also: Entry 1.4 (vital signs); Entry 2.5 (anaphylaxis); Entry 4.2 (bleeding); Entry 5.4 (sepsis).

Illustrative Case

ILLUSTRATIVE CASE: NOT A REAL PATIENT

Nadia worked the front desk at a community gym and was also a first-year EMT student. One evening a regular member, Frank, a man in his fifties, sat down on a bench after a warm-up and said he felt a pressure in his chest and a bit sick. He looked gray and was sweating. Nadia asked him to stay seated, called the emergency number, and asked a colleague to bring the AED from the wall near the desk.

While she waited, she checked his breathing and his pulse and kept him calm. A few minutes later, Frank slumped sideways and stopped answering. Nadia lowered him to the floor, shouted his name, and saw that he was not breathing normally; he gave an occasional gasp. She called to another member to tell the dispatcher that Frank had collapsed and was not breathing normally, and began chest compressions at the center of the chest, pressing hard and counting aloud to keep the pace.

The colleague returned with the AED. Nadia kept pressing while he opened it and placed the pads as the picture showed. The machine said it was analyzing the rhythm and told everyone to stand clear. Nadia lifted her hands. It advised a shock, and the colleague pressed the button after they all checked they were clear. Nadia resumed compressions at once. At the next rhythm check, Frank gave a deep breath. When the paramedics arrived, Nadia handed over what she had seen and the times.

Care Priorities

CARE PRIORITIES

When two cardiac problems compete, rank them by how soon each could end in an arrest. The patient who is unresponsive and not breathing normally comes first, ahead of every question you were about to ask. Next comes the awake patient whose chest pain is changing, because the trend decides how urgent the call becomes. Behind both stands the patient whose numbers look acceptable but whose skin, color or thinking has changed; trust that change over the number. Before a shift begins, find out where the nearest AED is kept and who is expected to fetch it, because a search made during an arrest spends minutes that compressions need. When you hand over, give the time of the collapse and the time of the first compression and the first shock, so the next team can judge how long the heart has gone without help.

Review Questions

1. Which statement distinguishes a heart attack from cardiac arrest?
   - **A)** They are the same event with two names.
   - **B)** A heart attack always causes cardiac arrest.
   - **C)** In a heart attack the heart muscle is injured, while in cardiac arrest the heart has stopped pumping effectively.
   - **D)** Cardiac arrest only happens in older adults.

2 A man in his fifties has had pressure in the center of his chest for 10 minutes, with sweating and nausea. What is the best first action?

A) Call emergency services and keep him still and comfortable.
B) Wait to see whether the pain eases.
C) Have a family member drive him to the hospital.
D) Let him rest and recheck in an hour.

3 On a monitor, a flat line appears in a patient who is awake and talking. What is the most likely explanation?

A) Cardiac arrest.
B) A loose lead or electrode.
C) A normal rhythm.
D) A heart attack.

4 For an adult in cardiac arrest, what rate of chest compressions is recommended?

A) 60 to 80 a minute.
B) 80 to 100 a minute.
C) 120 to 160 a minute.
D) 100 to 120 a minute.

5 What should you do when an AED advises a shock?

A) Check that everyone is clear, then press the shock button.
B) Keep pressing on the chest so that no time is lost.
C) Turn the AED off and wait for the paramedics.
D) Remove the pads and start again.

6 Which finding is an early sign of shock?

A) A very high blood pressure.
B) A strong pulse with warm hands.
C) A fast heart rate with cool, pale skin.
D) A very slow breathing rate.

7 Which statement about blood pressure in shock is correct?

A) A normal blood pressure rules out shock.

B) Blood pressure falls first, before any other sign.

C) Blood pressure is the only reliable sign of shock.

D) Blood pressure may stay normal for a time as the body compensates.

8 Short case. You are visiting a neighbor at home when he says his chest feels tight. A moment later he slumps in his chair, does not answer when you shout, and only gasps now and then. An AED hangs in the hallway. List, in order, the first five things you would do and say why compressions come before anything else you could do.